[Total No. of Question -04]

[Total No. of Pages - 01]

G H Raisoni International Skill Tech University, Pune

F.Y. B. Tech Sem-II

CAE II- SUMMER 2026 (2025 Pattern)

(Time: - 01 Hour) **Engineering Physics (25UBSL1204)**

(Max. Marks: 20)

Instructions to the students:

i) Solve Q1 or Q2 and Q3 or Q4. ii) Neat diagrams must be drawn wherever necessary. iii) Figures to the right indicate full marks. iv) Use of scientific calculator is allowed v) Assume suitable data, if necessary.

Q. No	*Question*	Marks	CO	BL
Q1.(a)	Explain the important characteristics of laser light.	[5]	2	2
(b)	Discuss the process of stimulated emission using an energy level diagram and explain its importance in laser operation.	[5]	2	2
	OR			
Q 2.(a)	Explain the main components of a laser with a neat diagram.	[5]	2	2
(b)	Describe the construction and working principle of a He Ne laser.	[5]	2	2
Q 3.(a)	Illustrate the forward and reverse bias operation of a PN junction diode using energy band diagrams and explain its relevance in semiconductor devices.	[5]	3	3
(b)	Discuss the formation of depletion region and potential barrier in a PN junction diode.	[5]	2	2
	OR			
Q4. (a)	Demonstrate optoelectronics and classify their different types with suitable examples.	[5]	3	3
(b)	Explain the construction and working principle of a Light Emitting Diode (LED) with a neat diagram.	[5]	2	2